

An introduction to measure theory - T.Tao

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Preface

Excercise 0.0.1. If $(x_\alpha)_{\alpha \in A}$ is a collection of numbers $x_\alpha \in [0, +\infty]$ such that $\sum_{\alpha \in A} x_\alpha < \infty$, show that $x_\alpha \neq 0$ for countable many $\alpha \in A$.

Intuition:

- Which also means that the number of positive x_α is countable.
- Divide and conquer: We can split original set (a.k.a scope) to smaller subsets and prove the statement locally to each subset before making out the point. 'make sure that the statement is valid on union operation'
- Countable means a set has finite elements, whereas uncountable set has infinite elements.

Proof. Because $\sum_{\alpha \in A} x_\alpha$ is convergent over A, it suffices to show that the sum must be convergent for every subsets of A. (1)

We split A into (can be overlapping cause that doesnt harm anything) subsets. Let n be a natural integer, A_n be subset of A, which includes indexes such that $x_\alpha \geq \frac{1}{n} \quad \forall \alpha \in A_n$, it is obvious that A_n has no zero elements. We first ensure that $\bigcup_{n \in \mathbb{N}} A_n = A$. That is, for every positive real x_α , we can always find large enough $n = \left\lceil \frac{1}{x_\alpha} \right\rceil \in \mathbb{N}$ that $x_\alpha \geq \frac{1}{n}$, thus $\exists n \in \mathbb{N}, x_\alpha \in A_n \quad \forall \alpha \in A, x_\alpha > 0$. As a result, we have **countably** infinite number of subsets. This extension is important because countability is preserved under union of countable number of sets.

We then prove by contradiction over these sets. Assume that there exists a natural number n such that subset A_n has infinite number of elements $||A_n|| = \infty$. Based on definition of A_n , we have $\sum_{\alpha \in A_n} x_\alpha \geq \frac{||A_n||}{n} = \infty$. This contradicts with (1). Therefore, for every natural number n, A_n has finite number of elements.

Finally, we conclude that the number of positive x_α is countable because $\bigcup_{n \in \mathbb{N}} A_n = A$ and each A_n has finite number of elements. Though it possible that we can prove the statement by using supremum over n, this conclusion is more concrete. $\square$

Excercise 0.0.2. *Exercise 0.0.2*

Chapter 1

Measure Theory

1.1 Prologue: The problem of measure

Exercise 1.1.1. *Exercise 1.1.1*